Deep Learning for Mineral and Biogenic Oil Slick Classification With Airborne Synthetic Aperture Radar Data

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***Abstract*— Studies of oil slicks in the ocean environment with synthetic aperture radar (SAR) have found that one of the most complex challenges to oil spill detection is the separation of mineral oil spills from slicks that are biogenic in origin. The possible occurrence of multiple scattering mechanisms beyond Bragg scattering for the sea surface, with or without biogenic or mineral oil slicks, and even under low to moderate wind conditions, has also been a subject of debate because the measured signals from these radar-dark surfaces can be contaminated easily by noise. Therefore, the use of noise-uncontaminated data is required for oil spill study in order to avoid significant alteration in the measured radar backscatter, which can lead to misinterpretation and misclassification of the scattering mechanisms involved. To this end, this study uses uninhabited aerial vehicle SAR data, with a noise-equivalent sigma zero as low as −53 dB, to investigate slick classification within a deep learning framework in order to assess deep architectures' capabilities for providing a reliable and accurate three-state classifier capable of separating mineral oil films from biogenic slicks and from the clean sea. The study exploits parameters with sensitivity to the dielectric constant and ocean wave damping properties, and convolutional neural networks' (CNNs') capability for learning nonlinear features, shapes, and textural and statistical patterns, in order to obtain significant classification accuracy. Very high accuracy results have been achieved, with values up to 0.91, 0.94, 0.98, and 0.99 under the most probable real-world spill acquisition conditions.**

***Index Terms*— Classification, convolutional neural networks (CNNs), deep learning, oil spill, synthetic aperture radar (SAR).**

I. Introduction

BOTH public opinion and political institutions have increased awareness and sensitivity to marine environment hazards, leading to increasing demand for dependable and continuous maritime monitoring tools in the face of the steady occurrence of pollutant releases into the sea. In this context, mineral oil spills present a major threat, with potential damages involving both on-shore and sea environments, affecting wildlife habitats, breeding grounds, and coastal water quality, and eventually leading to human health concerns. Research has shown how oil spills can affect sea organisms' and birds' life cycles [1], even demonstrating DNA damage in mammals fed with contaminated bivalves [2] and toxicological effects on directly exposed humans [3], all emphasizing the value of oil spill monitoring.

Due to its widely known all-weather and day–night observation capabilities [4], synthetic aperture radar (SAR) is often cited as the most valuable remote sensing instrument for sea oil spills detection under low to moderate (i.e., <8 m/s) wind speed conditions [5] and for oil slicks transport modeling and tracking [6]. Oil slicks under normal sea conditions appear radar-dark, having lower normalized radar cross section (NRCS, or σ^0) than clean water, as a consequence of the oil-dampening effects on the capillary and gravity-capillary surface waves [4], [5], [7] that arise from a reduction in the surface tension at the air–liquid interface [8].

In addition to mineral oil slicks, various other phenomena have been reported to cause noticeable NRCS reductions and are consequently referred to as oil spill look-alikes [4]. These include biogenic slicks (e.g., organic matter produced by vegetation in the intertidal zone and marine plants and animals), rain cells, low-wind and wind-shadow areas, some types of sea ice, zones of upwelling currents, microconvective cells, internal waves, ocean currents, and atmospheric or oceanic fronts [4], [5], [9]–[11]. Because of their similar and low NRCS values, separation of oil spills from look-alikes with high confidence level is a complex challenge to oil spill identification [5], and among the look-alikes, the greatest challenge is in differentiating mineral oil spills from biogenic slicks [7].

Several algorithms have been studied for oil classification with SAR, first using single-band/single-polarization sensors and the first- and second-order metrics [12], and subsequently taking advantage of polarization-dependent features from multipolarization and polarimetric [often called Polarimetric Synthetic Aperture Radar (PolSAR)] and multifrequency SAR sensors [4], [7], [8], [13]–[15]. The main concept underlying a possible improvement in oil spills discrimination with the use

Manuscript received May 21, 2020; revised August 30, 2020 and September 29, 2020; accepted October 20, 2020. Date of publication November 17, 2020; date of current version September 27, 2021. This work was supported by the Jet Propulsion Laboratory, California Institute of Technology, in contract with the U.S. National Aeronautics and Space Administration (Task Order) under Grant NNN13D788T. *(Corresponding author: Leonardo De Laurentiis.)*

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Digital Object Identifier 10.1109/TGRS.2020.3034722

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of multipolarization information is based on the assumptions that increased sensitivity to thick films is expected and that scattering mechanisms different from Bragg scattering take place in mineral oil slicks, with or without mixing with the seawater and even in the presence of weak wind fields. An often-made claim is that microwave backscattering from water and look-alikes can be modeled as quasi-deterministic Bragg surface scattering, whereas oil gives rise to surface scattering with a higher degree of randomness (higher entropy) and non-Bragg components, hence legitimizing the use of sensors with multipolarization capabilities for slicks discrimination based on the fact that PolSAR can be used to detect deviations from Bragg scattering [16].

Convincing arguments against the occurrence of non-Bragg scattering mechanisms for oil spills at sea and under low to moderate wind conditions have been published [7]. In this regard, studies using quad-polarimetric L-band data acquired with the airborne uninhabited aerial vehicle SAR (UAVSAR), from the NASA Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA, have confirmed that polarimetric decomposition features, such as H/A/$\overline{\alpha}$, prove to be consistent with a tilted Bragg scattering model [8]. Non-Bragg scattering mechanisms are consequently observable only in the event of moderate to high wind speed [17]–[19]. The inferred non-Bragg behavior can be attributed to noise-contaminated data due to the fact that spaceborne SARs typically have significantly higher noise floor than UAVSAR, which has noise-equivalent sigma zero as low as -53 dB [8], [20]. In fact, low backscatter signals from look-alikes and slicks that are below a spaceborne SAR's noise floor [21] can cause errors and misinterpretation. Thus, the use of sensors with ultralow-noise floors, such as UAVSAR, is definitely desirable in monitoring and discriminating oil slicks in order to avoid measurement errors and slick/slick-free surface misclassification [22] due to noise-contaminated data [23], [24].

Several articles dealing with SAR oil slick classification have been published, including some that make use of deep learning architectures and the less complex neural networks. Many authors claim high to very-high accuracy scores [25]–[30] achieved on RADARSAT-2 and ERS-2 data (87%–98.9%) by harnessing both shallow and deep neural networks. However, the studies set a two-class problem (mineral oil versus clean-sea, mineral oil versus clean-sea and/or biogenic slicks, and mineral oil versus look-alikes) instead of three classes (mineral oil versus biogenic slicks versus clean-sea) or more, and rely on polarimetric features and data that may be corrupted by the aforementioned noise contamination. An interesting ten-class deep classification with SAR Sentinel-1 data is presented in [9], with valuable overall accuracy [see (11)] results but lower scores for different metrics, such as the recall [see (12)]; however, we only consider biogenic slicks rather than mineral oil slicks, and in addition, all the aforementioned noise considerations apply. An interesting five-class semantic segmentation with SAR Radarsat-2 and SIR-C/X data is given in [31], obtaining a 90.5% accuracy. However, the use of a relatively small data set (three Radarsat-2 and SIR-C/X SAR images and 478 samples), in addition to potential noise contamination, leaves this open to further studies.

In the work presented here, an assessment of deep neural networks' capabilities in performing oil slick classification is discussed as a three-class (mineral oil versus biogenic slicks versus clean-sea) classification problem. Through the use of a large UAVSAR data set (54 SAR images), noise contamination problems are significantly reduced due to the low-noise floor. Furthermore, to avoid bias from changing meteorological conditions or wind speeds not conducive to the slick biogenic formation, all data in the employed data set are collected under low to moderate wind speed conditions. It is noteworthy that specific input data have been selected to exploit radar's expected sensitivity to oil damping effects, to large-scale and small-scale roughness, and to the effective dielectric constant of surface films based on scattering theory and prior studies, such as [32], [33], which have already shown that features independent of small-scale roughness (the ocean wave spectrum) result in poor slicks separability, while features dependent on the spectrum have higher sensitivity; among the best small-scale roughness-dependent features, the vertical transmit and vertical receive (VV) damping ratio and the polarization difference (PD) have proven very high-performance [32], [33]. The choice of a parameter sensitive to the dielectric constant follows the work of Minchew *et al.* [8] and [34] and its use for discriminating between mineral and biogenic slicks is supported by the fact that SAR's sensitivity to the dielectric properties of the surface layer is expected to be different for mineral oil and biogenic slicks due to their different hydrophobic characteristics [7]. Furthermore, Angelliaume *et al.* [35] showed that the dielectric coefficient difference between mineral oil films and seawater/biogenic films is reduced for increasing radar frequency, making L-band data more suitable for dielectric-based distinction purposes than C-, X-, and K-bands. SAR sensitivity to slicks' constituents leads to differences in the radar backscattering statistical behavior [4], [7], [36], potentially observable in SAR products. Recent deep convolutional architectures have shown the capability to deal with shapes and textural features at many levels of detail, making them more effective than pixel-based approaches, even those using optimal parameters. They are particularly appropriate for oil film discrimination given that, quite often, different spreading mechanisms occur for different types of films, contributing to their forming distinctive spatial patterns on the sea surface.

Theoretical aspects will be further elaborated in Section II, followed by data analysis in Section III. As shown in Section IV, very high overall accuracy results have been achieved, together with very high scores for different evaluation metrics, such as precision, recall, and F-1 score, thus proving the expected effectiveness of deep architectures in this context. We end with a discussion on the ideal hyperparameters setting in a possible operational scenario.

II. Theoretical Background

A. Physics Framework

Ocean surface backscattering is generally described in terms of the Bragg scattering theory, in which the radar's electromagnetic wave is in resonance with ocean surface waves and, to second order, can undergo a supplementary modulation due

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to the longer gravity waves [37]; this behavior is described theoretically by the tilted Bragg model (also called composite-surface model, two-scale Bragg scattering theory, and Valenzuela model) [7], [8], [32], [38]. Ocean roughness features mainly contribute to the generation of co-polarized backscatter, related to the presence of ocean waves with wavenumber equivalent to

$$k_B = 2k_r \sin\theta_i \tag{1}$$

where k_r is the radar wavenumber and θ_i is the incidence angle [39]. In the tow-scale Bragg scattering theory, cross-polarized backscatter is attributable to a tilt due to gravity waves, with an incidence angle equal to

$$\theta_i = \cos^{-1}[\cos(\theta+\psi)\cos(\zeta)] \tag{2}$$

where ψ is the angle between the vertical and the normal projection into the incidence plane, ζ is the angle between the vertical and the normal projection into the plane perpendicular to the incidence plane, and θ is the incidence angle relative to the untilted horizontal plane [8], [38]. Tilt occurs more often for range traveling waves than azimuth traveling waves, which are more affected by an additional doppler component referred to as velocity bunching [40]; the most notable consequences are apparent densification (also called bunching) and thinning of scatterers, together with a position shift in the resulting SAR image [40]). Based on these considerations, in the literature, it is commonly assumed that tilt modulation has its largest impact on the range direction [41], and it is often implied that $\zeta = 0$; consequently, the relevant geometry is frequently shown in 2-D, as in Figs. 1 and 2 [38], [40], [41].

The equations of the tilted Bragg model describing radar backscattering are written as

$$\sigma^0_{\mathrm{HH}} = 4\pi k_r^4 \cos^4(\theta_i) \left| \left(\frac{\sin(\theta+\psi)\cos(\zeta)}{\sin(\theta_i)}\right)^2 R_{\mathrm{HH}}(\theta_i,\epsilon_r) + \left(\frac{\sin(\zeta)}{\sin(\theta_i)}\right)^2 R_{\mathrm{VV}}(\theta_i,\epsilon_r)\right|^2 \times W(2k_r\sin(\theta+\psi), 2k_r\cos(\theta+\psi)\sin(\zeta)) \tag{3}$$

$$\sigma^0_{\mathrm{VV}} = 4\pi k_r^4 \cos^4(\theta_i) \left| \left(\frac{\sin(\theta+\psi)\cos(\zeta)}{\sin(\theta_i)}\right)^2 R_{\mathrm{VV}}(\theta_i,\epsilon_r) + \left(\frac{\sin(\zeta)}{\sin(\theta_i)}\right)^2 R_{\mathrm{HH}}(\theta_i,\epsilon_r)\right|^2 \times W(2k_r\sin(\theta+\psi), 2k_r\cos(\theta+\psi)\sin(\zeta)) \tag{4}$$

$$\sigma^0_{\mathrm{HV}} = 4\pi k_r^4 \cos^4(\theta_i) \left(\frac{\sin(\theta+\psi)\cos(\zeta)\sin(\zeta)}{\sin^2(\theta_i)}\right)^2 \times |R_{\mathrm{VV}}(\theta_i,\epsilon_r) - R_{\mathrm{HH}}(\theta_i,\epsilon_r)|^2 \times W(2k_r\sin(\theta+\psi), 2k_r\cos(\theta+\psi)\sin(\zeta)) \tag{5}$$

where $W()$ is the 2-D wavenumber spectral density of the ocean surface roughness, and the Bragg scattering coefficients

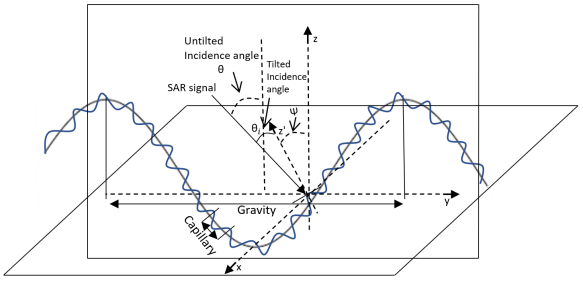


Fig. 1. Relevant geometry of tilted Bragg model in range, after [38] and [40]–[43].

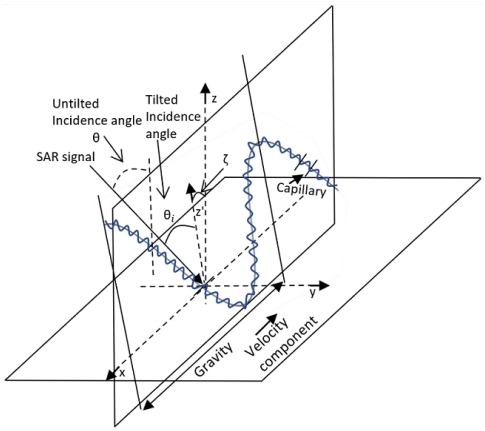


Fig. 2. Relevant geometry of tilted Bragg model in azimuth, after [38] and [40]–[43].

R_{ij} are modeled as

$$R_{\mathrm{HH}}(\theta_i,\epsilon_r) = \frac{\cos(\theta_i) - \sqrt{\epsilon_r - \sin^2(\theta_i)}}{\cos(\theta_i) + \sqrt{\epsilon_r - \sin^2(\theta_i)}} \tag{6}$$

$$R_{\mathrm{VV}}(\theta_i,\epsilon_r) = \frac{(\epsilon_r - 1)[\sin^2(\theta_i) - \epsilon_r(1+\sin^2(\theta_i))]}{(\epsilon_r\cos(\theta_i) + \sqrt{\epsilon_r - \sin^2(\theta_i)})} \tag{7}$$

where the relative complex dielectric constant is represented by ϵ_r [42], [43]. The backscattered NRCS is, thus, dependent on the ocean wave spectrum $W()$, on the dielectric coefficient values and on the geometry of acquisition. Furthermore, the ocean wave spectrum $W()$ is polarization-independent in the two-scale Bragg scattering theory and is canceled in the backscattering ratios of different polarizations [8], making them only related to the acquisition geometry and dielectric properties [23]. It is noteworthy that in the case of oil films or emulsions at sea, the NRCS reduction is due to two different mechanisms: the mineral oil damping effect on the ocean waves, and the ocean surface dielectric constant reduction due to the presence of the same mineral oil [7]. In this context, the co-polarized backscattering ratio provides a very simple method to evaluate dielectric constant variations

$$\frac{\sigma^0_{\mathrm{HH}}}{\sigma^0_{\mathrm{VV}}}. \tag{8}$$

For thin films on the ocean surface, the backscatter is mainly related to the dielectric properties of seawater (since transmitted microwaves penetrate the thin layer, reaching the

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underlying seawater, from which the microwaves backscatter), and therefore, the NRCS is more affected by dielectric properties in the case of thicker mineral oil films [7], [8]. In this regard, the use of the co-pol ratio [(8)] can be beneficial for mineral oil/biogenic slicks discrimination. Biogenic films can persist on the ocean surface only in the form of thin monomolecular layers (∼2.4–2.7 nm), while mineral oil films can persist as thicker multilayers (∼1 μm–1 mm) because of their different hydrophobic constituents. As a result, microwave radar instruments are not sensitive to dielectric properties of biogenic films [14]. Typical values of the real part (proportional to the dielectric polarization degree) and of the imaginary part (proportional to the dielectric loss factor) of the complex dielectric constant at L-band are $\sim 70 < \epsilon_r' <\sim 90$ and $\sim 45 < \epsilon_r'' <\sim 95$ for seawater [44]; furthermore, the dielectric constant of mineral oil at L-band (1.26 GHz) is approximately $\epsilon_r = 2.3 + 0.01i$ [45]. When oil becomes emulsified, the mixture of oil and water will increase the dielectric constant depending on the oil-water concentrations, $\sim 25 < \epsilon_r' <\sim 55$ and $\sim 25 < \epsilon_r'' <\sim 45$ given a typical 25%–50% oil concentration percentage in marine oil slicks [46]. On the other hand, ϵ_r values for (biogenic) monolayers are referred to as comparable to seawater [7], [8], [34], [44], [45], [47] because of the electromagnetic waves penetration through the film. The co-pol ratio is consequently expected to be similar for areas of clean-sea water and areas of water covered with biogenic slicks, whereas values for areas of thick mineral oil films could be modified by the surface film's dielectric properties.

In general, for most spaceborne SAR instruments, NRCS values for mineral oil films and look-alikes are very close to the instrument noise floor. Moreover, cross-pol measurements are more sensitive to corruption from instrument noise as compared with co-pol measurements due to their generally lower values. Hence, methods that do not use cross-polarization data are generally less susceptible to instrument noise. Finally, L-band has been observed to be more suitable for dielectric-based differentiation [35], and biogenic films exhibit more NRCS suppression than mineral oil spills only at L-band [15], [48].

The inferred non-Bragg scattering from mineral oil films is observable in moderate to high wind speed conditions, and therefore, cannot be used to discriminate between mineral oil films and biogenic slicks, at least under low to moderate conditions. Under moderate to high wind speed conditions (i.e., >8 m/s), non-Bragg scattering processes contribute to radar backscattering due to the steepening of the waves and wave breaking [17], [18]. Low to moderate conditions are needed to study slicks since higher winds may disperse and mix the slicks into the upper ocean unless the oil is emulsified and weathering has occurred. For this reason, slick discrimination with SAR and under high wind speed conditions has been reported as not being possible [15] (although more recently observed up to 12 m/s [49]). More rapid formation of mineral oil emulsions is observed with wind speed >7 m/s [7], while biogenic slicks are reported to experience washing-down and entrainment effects with wind speed >7–8 m/s [50], [51]. Moreover, organic biogenic slicks were reported to be observed only at wind speeds <5 m/s based on data from the Red Sea and Gulf of Aden [52].

The data set employed for the work reported here, entirely based on UAVSAR L-band products, has been selected with special care to meet the requirements for homogeneous, low to moderate wind speed conditions, as is further discussed in Section II-B. The use of UAVSAR L-band allows the exploitation of high-resolution data, combined with the aforementioned advantages regarding the use of a unique sensor with an extremely low-noise floor and dielectric properties sensitivity. In addition, UAVSAR has shown accurate oil slick detection even in high wind speed conditions [49]. Although Skrunes *et al.* [49] found that dielectric-based or general oil films/biogenic slicks discrimination was not feasible, clear differences in shapes between the biogenic and mineral oil slicks were observed.

From an operational point of view, unlike a satellite-borne SAR sensor, an airborne instrument, such as UAVSAR, can enable rapid-repeat imaging, which is desirable for monitoring slick short-term changes. Moreover, recent results [32], [35], [53], [54] prove the high reliability of a UAVSAR-derived metric, namely, the VV damping ratio, defined in [53] as the contrast between the VV-intensity in clean versus slicked water

$$\zeta_{\mathrm{VV}} = \frac{\mathrm{VV}^{\mathrm{clean}}}{\mathrm{VV}} \tag{9}$$

which is shown to be a reliable parameter for detecting/characterizing oil slicks and for separating biogenic slicks from clean water and mineral oil films from clean water, e.g., based on the Jeffries–Matusita (JM) distance metric in [32] (JM metric is a function of statistical parameters, used for evaluation of two classes separability; usually JM $\geq$ 1 is an indication of separability). As a general principle, oil films spread over the water surface, forming a layer that damps small-scale gravity-capillary waves [32], [55]; thicker layers give rise to stronger damping [53], motivating the idea that the damping ratio might prove a practicable method to differentiate types of slicks. The damping ratio, defined as such [i.e., (9)], is also a practical parameter because it requires only a single polarization, reducing instrument complexity [53], and proves to be an effective and convenient way to measure slick damping effects. Moreover, VV polarization is preferred to the HH and HV polarizations for the damping ratio feature since the VV channel has proven higher slick/clean sea contrast and higher signal-to-noise ratio [33], [51], [56]–[58].

Further results described by [32] and [33] indicate that another small-scale roughness-dependent feature is of great relevance for both discrimination purposes and for quantifying oil-water relative concentration, namely, the co- PD defined as [33]

$$\mathrm{PD} = \sigma^0_{\mathrm{VV}} - \sigma^0_{\mathrm{HH}} \tag{10}$$

PD is characterized by sensitivity to the short wind Bragg wave contribution and the capacity of revealing the slick concentration variation at sea, demonstrated with L-band airborne SAR [33]. A variety of methods to distinguish biogenic slicks and mineral oil patches that are not exclusively based on the reduction in NRCS values have been presented in the literature.

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Differentiation is possible, under certain conditions, even with the position- and shapes-based algorithms, together with methods based on differences in the image statistics and textural information. A few remarks are of interest in this context: wind can exert a stronger force on heavier oil components, which gives the downwind edge of mineral oil slicks a sharp boundary; however, this condition does not occur in monomolecular biogenic slicks. Furthermore, mineral oil films often exhibit recurring features when in the same area [59], [60]. Similarly, biogenic slicks may show specific patterns, e.g., the streaklines present in the Wax Lake Delta (WLD) area [10], [51], [59], [61] regardless of wind speed. Skrunes *et al.* [36] gave a different analysis, based on intensity statistics and found that the NRCS of mineral oil appears to be more spatially inhomogeneous relative to the NRCS of biogenic slicks. Similar and related conclusions are reached in [4], [24], and [62], relying on the co-polarized phase difference statistics. Nevertheless, the co-polarized phase difference feature (also abbreviated as CPD) and the standard deviation (StdDev) of the CPD are not considered in this study because they have been found to be very similar to the CoPolRatio at L-band [32], [49]. Several studies have pointed out the value of the StdDev of CPD in oil slick detection at L-band [63], [64], while others suggest that the CPD approach is not beneficial for oil slicks classification at L-band. In particular, Migliaccio *et al.* [4] demonstrated results consistent with those reported by Gade *et al.* [15], emphasizing that the CPD approach is not beneficial for oil spill discrimination at L-band. Moreover, Minchew *et al.* [8] found that the StdDev of CPD at L-band exhibits virtually no difference for oil and water except in the presence of instrument noise. This is consistent with theoretical considerations well explained in [4], [8], [63], and [64], showing that weak-damping slicks (e.g., monomolecular biogenic slicks) are indistinguishable from the clean sea using the StdDev of CPD.

B. Deep Learning Framework

In view of the above-mentioned references, the idea of assessing machine/deep learning performances in oil slick classification is straightforward. One of the main strengths in common between machine learning architectures is represented by the statistical estimation potential. Feature learning and inference based on data statistics are among the most recognizable capabilities of these architectures [65], [66]. Moreover, recent deep architectures, such as convolutional neural networks (often shortened as ConvNets or CNNs), showed their strength in a variety of tasks. Among the most interesting applications, it is worth mentioning: semantic segmentation of images and action/objects recognition in movies [67], texture analysis [68], facial emotion recognition [69], and a wide range of applications in remote sensing, such as automated ship detection from aerial images [70], SAR ship classification [71], remote sensing image classification [72], and development of enhancement methods for remote sensing observations (for a review, see [73]). CNNs are capable of taking advantage of textural features in the underlying data, along with information content at different levels. Deeper layers, with the increase of receptive fields of neurons, complexity, and level of abstraction, create a richer semantic representation in a hierarchical way [74], allowing low-to-high level feature extraction. Through the use of convolutional kernels and pooling operations, CNNs obtain key properties, such as sparse connectivity, parameter sharing, and translational invariance [65], while the use of activation functions, such as the rectified linear unit, provides the capability to learn nonlinear features from underlying data. Different arrangements of smaller layers in the form of modules paved the way for the exploration of extreme representations at lower scales, stacked together to form more powerful networks, with proven better performances (e.g., within the ImageNet Large Scale Visual Recognition Challenge [75]), such as with the introduction of inception [76], xception (XCT) [77], and residual blocks [78].

Building on the same line of research, a relevant goal of the work presented here is the evaluation of deep CNNs' performances in conducting the three-class problem of classifying mineral oil slicks versus biogenic slicks versus clean seawater with SAR data. The use of the UAVSAR sensor supports a noise-uncontaminated, high spatial resolution analysis, and for these reasons, this sensor is a recommended choice, as is also the similar F-SAR airborne SAR system [7], [79]. Furthermore, the use of the co-polarized backscattering ratio, the VV damping ratio, and the PD [see (8)–(10)] provides the means to account for differences in both the dielectric properties and ocean wave spectra. Tools that use only pixel-level information are challenged in differentiating low-level signals, especially using satellite SARs, but other information is available that can inform classification; the application of CNNs provides a unique tool that combines pixel-level values with spatial features, shapes, and textural properties, and enable learning based upon statistical properties. This is expected to definitely improve classification, given that even if different scattering mechanisms are not present, different statistical patterns should be observed [7] due to differences in the homogeneity and thickness of mineral and biogenic oil slicks.

In view of the above, an assessment of two different deep CNN architectures within the three-class mineral oil slick versus biogenic slick versus clean seawater classification problem is carried out in the present work. Chosen architectures are known as XCT [77] and InceptionResNet (IRN) [80], already proven to have best performances on an SAR-domain classification (ship classification) with CNNs [71]. XCT consists of 36 Conv-layers, with unique arrangements and some of them reoccurring, while IRN consists of 467 layers. Both architectures combine the use of convolutional and pooling inception-based blocks with residual connections (for further architectural details, see [77], [80]). Different performance parameters have been assessed, with the use of identical hyperparameters, no data augmentation, and the same downstream fully connected NN (FCNN) structure that drives the final classification; this latter has been set in a "small configuration" of a 32-16-3 three-layers structure, proven better with classification over the complete data set in [71]. A complete representative diagram is shown in Fig. 3; it is noteworthy that each complete architecture holds its own FCNN, and

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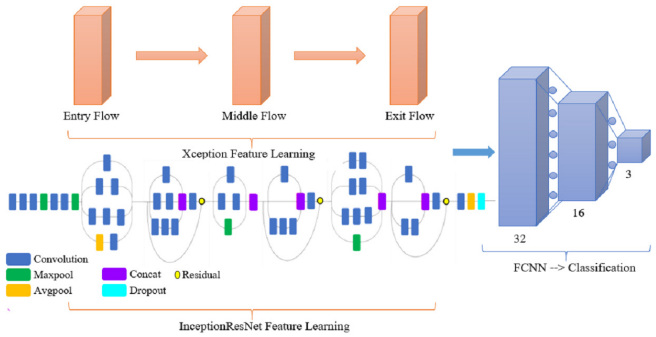


Fig. 3. Schematic of involved CNNs and related Classification. XCT feature learning architecture is based on the entry-middle-exit flows in [77]. IRN feature learning architecture is based on [76] (part of this figure is based on [81]).

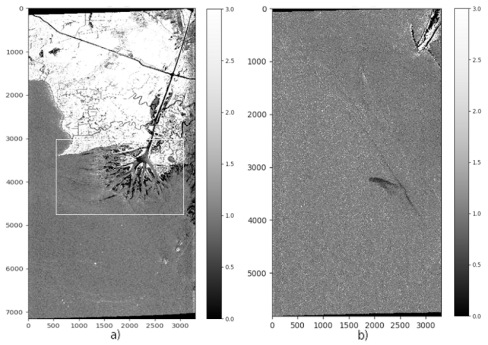


Fig. 4. UAVSAR VV acquisitions over the WLD area in the American state of Louisiana: (a) Flight ID-Flight Line: 15057-005, and over the part of the Mississippi Canyon Block 20 in the Gulf of Mexico containing the destroyed Taylor Energy platform 23051, also known as Taylor Rig or MC-20 area (b) Flight ID-Flight Line: 16099-004. An enlarged view of the area within the box in (a) is shown in Fig. 5.

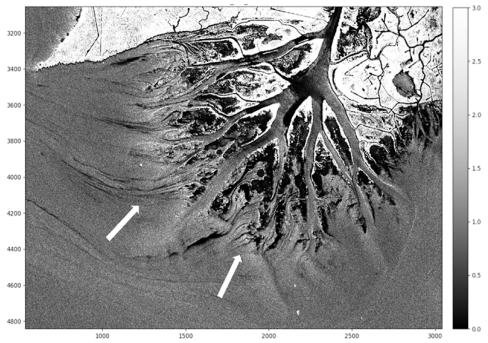


Fig. 5. Enlarged view of UAVSAR VV acquisition over the WLD area. Biogenic slicks samples are indicated by white arrows.

Fig. 3 only emphasizes that the same FCNN design has been set for all the experiments. The complete set of data has been provided in the form of patches, which represent the basic units for CNNs training, validating, and testing. From the operational point of view, oil occurrence percentage (percentage of pixels containing slick) and patch size search-spaces are explored in order to find the best parameters for a range of real-work conditions.

III. Data Analysis

A. Study Sites Details

The UAVSAR sensor operates at L-band (1.2575-GHz center frequency) and with full-polarization capabilities [53], and various standard products are available. The calibrated multilooked complex (MLC) products have been selected for the present study, which is given with three range looks, 12 azimuth looks, 5-m pixel spacing in slant range, and 7.2-m pixel spacing in azimuth [8]. Since the experimental data inputs used in this study are the VV damping ratio, co-polarized difference, and co-polarized backscattering ratio [see (8)–(10)], only HH- and VV-polarization products are of interest. The data set used is based on UAVSAR acquisitions carried out over two different areas of interest, namely, the WLD area in the American state of Louisiana [10], and the part of the Mississippi Canyon Block 20 in the Gulf of Mexico containing the destroyed Taylor Energy platform 23051 (also known as Taylor Rig or MC-20 area) [53], [82]. Sample UAVSAR VV acquisitions over the two areas are shown in Figs. 4 and 5. The Taylor drilling platform, located about 16-km southeast of the end of the Mississippi River (Gulf of Mexico), was severely damaged by Hurricane Ivan in 2004 [83]. Since then, the area has been the site of frequently observed mineral oil slicks [53], [83], with detected slicks as thick as 1–2 mm [53], [82], [84]. The optimum wind speeds to study oil films in this area has been reported to be in the range from 3.5 to 7.0 m/s [84]. The Wax Lake Delta is a small, actively prograding delta near the west edge of the Mississippi River watershed [10]. A comprehensive study on wetland vegetation and biomass estimation in the WLD area is given in [85]; multiple acquisitions within the same area revealed evidence of streaklines that are likely composed of biogenic slicks that accumulate in the Delta area [10], [86], a general class of biogenic slicks discussed in other marine environments [50], [51], [61]. Observations given in [10] indicate that streaklines observed in this area originate independently of wind fields; consequently, wind speed is not important for the creation and observation of biogenic slicks in the Wax Lake Delta provided that values remain below the moderate threshold on the Beaufort scale. However, measured wind speeds for products in the data set used never exceed the limit referred by [52], which claimed that biogenic slicks should only be detectable for wind speeds <5 m/s.

UAVSAR acquisitions over MC-20 mineral oil slicks and the Wax Lake Delta were selected to constitute mineral oil and biogenic slicks patches in the custom data set employed in the present work, respectively. Areas of clean-sea water have been sampled within the same areas and incidence angle range in order to assure the same imaging conditions. A complete list of selected products is given in Table IV: 22 UAVSAR MLC products have been chosen for the MC-20 area, and 32 UAVSAR MLC products have been chosen for the WLD area. All selected acquisitions comply with aforementioned wind speed requirements, verified using data from the National Data Buoy Center, National Oceanic and Atmospheric Administration, Washington, DC, USA, which are likewise summarized in Table IV. Coincident wind data from AMRL1 station (29°26'58"N and 91°20'17"W), BURL1

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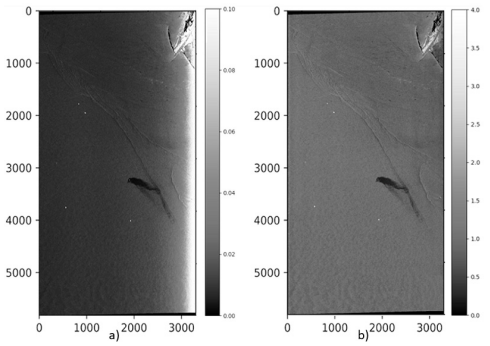


Fig. 6. Correction procedure application example, performed on UAVSAR data acquired on November 15, 2016 (Flight ID-Flight Line: 16099-008). The procedure is described in detail in [32] and is based on an empirical estimation of a correction factor by considering a homogeneous clean ocean region.

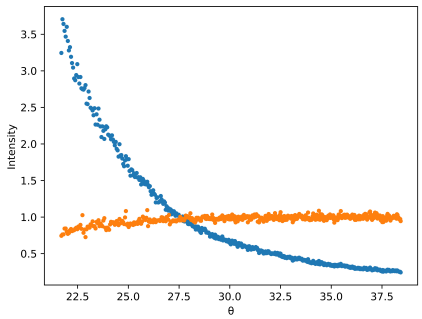


Fig. 7. Intensity profile versus incidence angle θ before (blue-dots) and after (orange dots) applying the correction procedure.

station (28°54'18"N and 89°25'42"W) and LLNR-245 station (29°12'30"N and 88°13'33"W) were obtained to confirm requisite conditions in the Wax Lake Delta (AMRL1) and MC-20 (BURL1, LLNR-245) areas.

UAVSAR data are generally acquired within an incidence angle range of approximately 20°–65°; a comprehensive list of the sensor parameters is given in Table I. Backscatter values from ocean decreases with increasing incidence angle, and thus, intensity variations due to varying incidence angles may lead to major differences between different classes and within the same class. In order to minimize incidence angle effects on the classification, the incidence angle correction algorithm given in [32], based on empirical estimation of a correction factor by considering a homogeneous clean ocean region, has been exploited. Correction of incidence angle effects has been carried out as the first step in the presented assessment, and an illustration of incidence angle effects, together with an intensity profile after correction, are given in Figs. 6 and 7. As observed in [32], corrected intensity profiles are not completely flat due to some local variations.

B. Experimental Data Details

1) Hyperparameters: In the present assessment, a grid-search process is set in order to find the best hyperparameters from an operational point of view; in particular, the investigation is focused on patches size and oil-occurrence-percentages

TABLE I
UAVSAR PARAMETERS

Parameter	Range of Values
Frequency	1.2575 GHz (0.2379 m)
Bandwidth	80 MHz
Polarization	Quad-Pol (HH and VV channels have been considered)
Range Swath	20 km
Incidence Angle	20°- 65°
Antenna Size	0.5 m x 1.6 m
Nominal Altitude	12.5 km
Operating Altitude	2-18 km

TABLE II
DATA SET OVERVIEW. WLD AREA AND MC20 FOR THE MC-20 TAYLOR RIG AREA

Side Size (pxls)	Oil Percentage	Numbers	Class	Area
128, 256	10%	23288	Biogenic slick	WLD
128, 256	30%	10840	Biogenic slick	WLD
128, 256	50%	5290	Biogenic slick	WLD
128, 256	70%	2252	Biogenic slick	WLD
128, 256	10%	23288	Mineral oil	MC20
128, 256	30%	10840	Mineral oil	MC20
128, 256	50%	5290	Mineral oil	MC20
128, 256	70%	2252	Mineral oil	MC20
128, 256		11644	Clean-sea water	WLD
128, 256		11644	Clean-sea water	MC20

TABLE III
OVERVIEW OF TRAINING, VALIDATION, AND TEST SETS' SIZES

Oil Percentage	Training Set	Validation Set	Test Set
10%	44712	11180	13972
30%	20812	5204	6504
50%	10156	2540	3174
70%	4324	1080	1352

parameters. The search spaces consist of two different patches size (128 and 256 pixels on each side, respectively) and four different minimum oil occurrence percentage values (from 10% to 70%, with a step of 20%); in the latter, percentage values indicate the minimum percentage of pixels belonging to the oil class (same considerations apply both for biogenic oil and mineral oil films) that are present in the given patch, and hence, 10% patches may contain from 10% to higher oil-occurrence percentages, and so on for other percentage values; a sample illustration is shown in Fig. 8, where two 10% and 70% possible extreme cases are reported. The complete data set is, therefore, made up of patches of different size and different oil percentage values; a data set overview is given in Table II, while Table III specifies training, test, and validation sets details of the final data set, on which CNNs are trained, validated, and tested.

2) Data Set Creation: Data set construction, from which training, validation, and test sets are derived, has been carried out in an automated way, based on a set of previously designed mineral/biogenic oil and water masks. These latter have been manually derived by expert visual inspection over the whole set of UAVSAR products listed in Table IV. For each class, given a fixed oil-occurrence percentage, the number of samples has been dimensioned according to the less-represented class, random sampling the remaining classes in order to get a balanced data set for each oil-occurrence-percentage; this method is known as random undersampling (RUS), which showed good performances and mostly outperformed random over-sampling and intelligent sampling techniques in deep learning with class imbalance [87]. Imbalance effects represent a known issue in machine learning, due to which biased

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TABLE IV
UAVSAR L-BAND PRODUCTS

Date	Time	Flight ID	Flight Line(s)	Wind Speed (m/s)	Area
8-May-2015	12:29:12 UTC	15057	005	1.8-2.7	Wax Lake Delta
8-May-2015	12:56:50 UTC	15057	007	1.7-2.2	Wax Lake Delta
8-May-2015	11:20:46 UTC - 14:09:07 UTC	15057	000, 002, 004, 010, 012	1.3-2.8	Wax Lake Delta
8-May-2015	12:43:11 UTC	15057	006	1.6-1.7	Wax Lake Delta
8-May-2015	13:10:33 UTC	15057	008	2.4	Wax Lake Delta
12-Aug-2019	13:15:50 UTC	19053	019	2.3-2.7	Wax Lake Delta
16-Oct-2016	13:52:58 UTC - 15:52:34 UTC	16089	000, 004, 008	1.5-3.2	Wax Lake Delta
17-Oct-2016	21:08:16 UTC - 22:05:10 UTC	16090	008, 012	2.5-3.2	Wax Lake Delta
17-Jun-2009	18:22:22 UTC	09045	006	2.0-2.1	Wax Lake Delta
02-Sep-2009	19:15:12 UTC	09066	009	2.4-2.9	Wax Lake Delta
25-Jan-2010	21:16:48 UTC	10009	003	1.9-3.0	Wax Lake Delta
23-Jun-2011	17:15:24 UTC	11038	008	1.6-1.9	Wax Lake Delta
02-Jul-2012	21:09:31 UTC	12053	015	0.0	Wax Lake Delta
25-Oct-2012	13:36:15 UTC	12114	006	1.4-1.8	Wax Lake Delta
29-Oct-2013	18:46:47 UTC	13163	008	5.0	Wax Lake Delta
09-Apr-2014	21:04:44 UTC	14036	012	2.3-2.7	Wax Lake Delta
28-Oct-2014	19:42:15 UTC	14161	010	1.6-2.3	Wax Lake Delta
06-May-2015	04:40:38 UTC	15056	005	3.8-4.2	Wax Lake Delta
09-May-2015	16:01:55 UTC	15058	004	3.1-4.2	Wax Lake Delta
06-May-2015	04:15:32 UTC	15056	004	4.3-4.9	Wax Lake Delta
08-May-2015	11:34:05 UTC - 13:53:50 UTC	15057	001, 003, 009, 011	1.6-2.4	Wax Lake Delta
17-Oct-2016	22:28:19 UTC	16090	013	0.1-2.6	Wax Lake Delta
08-May-2015	14:38:32 UTC	15057	013	3.3-5.4	MC20 Taylor Rig
15-Nov-2016	12:12:04 UTC - 14:25:02 UTC	16099	001, 002, 003, 004, 005, 006, 007, 008	4.2-6.7	MC20 Taylor Rig
17-Nov-2016	15:31:16 UTC - 17:47:53 UTC	16100	001, 002, 003, 004, 005, 007, 008	3.8-4.9	MC20 Taylor Rig
17-Nov-2016	21:58:04 UTC - 23:47:28 UTC	16101	000, 001, 002, 003, 004, 005	4.3-6.2	MC20 Taylor Rig

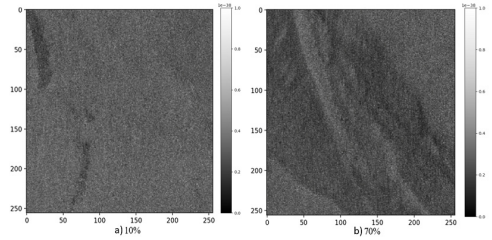


Fig. 8. Sample illustration of visual differences between two mineral oil-occurrence extreme cases: a 10%-oil patch (on the left-hand side) and a 70%-oil patch (on the right-hand side), shown here for the MC20 co-polarized backscattering ratio. The oil-percentage is a lower bound, i.e., 10%-oil means a minimum of 10%. The two extreme cases are shown to summarize and clearly emphasize the possible endpoints gap, with the other two cases falling in between (30% and 50%).

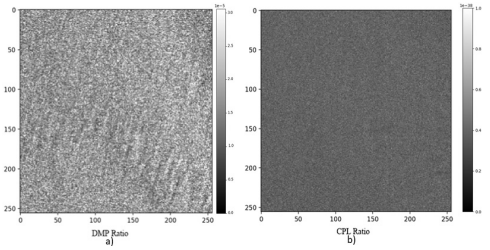


Fig. 9. Illustration of visual differences between VV damping ratio (DMP ratio, left-hand side) and co-polarized backscattering ratio (CPL ratio, right-hand side) for the clean-sea-water class.

learning can take place. Severely imbalanced data sets lead to major difficulties, as several learners will reveal a significant bias toward the largest class, and in extreme situations, might completely not consider minority classes [87], and hence, RUS provides a straightforward method to give a balanced data set that allows the avoidance of biased-learning effects associated with unbalanced data sets. Besides, various evaluation metrics may lead to erroneous interpretations of performance indicators in case of imbalance side effects; this is the case for the overall accuracy metric. For a more detailed evaluation, it is common to give further metrics, e.g., precision, recall, and F-1 score [88]. Even though a balanced data set has been used in this work, the overall accuracy may not be sufficient to have a comprehensive view on performances, and thus, additional metrics are provided and further discussed in Section III-B.3.

3) Experimental Framework: Various experiments have been set, involving performance assessment of XCT and IRN architectures that can be conceptually divided into two classes: single- and double-input studies. Single-input experiments are meant to investigate CNNs performances when trained over a single-input type (VV damping ratio or co-polarized difference or co-polarized backscattering ratio), while double-input ones operate by simultaneously taking both VV damping ratio (the small-scale roughness-dependent feature with less high scores) and co-polarized backscattering ratio as the input of CNNs. This experiment is mainly to investigate whether the additional input of a small-scale roughness independent feature, with dielectric properties sensitivity, can improve the classification in a double-input scenario, as further explained in the following. Given this, details are given in Tables II and III refer to double-input data sets; to obtain values for the single-input case, one can simply halve the numbers. Furthermore, in order to have a visual illustration of input data, Figs. 9–11 are given to show key differences between input data products; VV damping ratio products (referred to as DMP ratio) are placed on the left-hand side in Fig. 9–11, and co-polarized backscattering ratio products (referred to as CPL ratio) are placed on the right side. The co-polarized difference is visually

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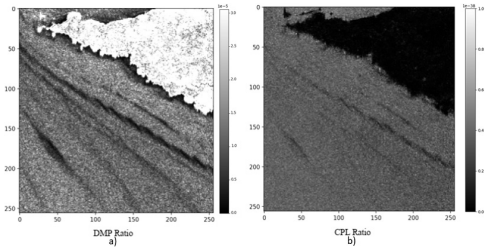


Fig. 10. Illustration of visual differences between VV damping ratio (DMP ratio, on the left-hand side) and co-polarized backscattering ratio (CPL Ratio, on the right-hand side) for the biogenic-slicks class.

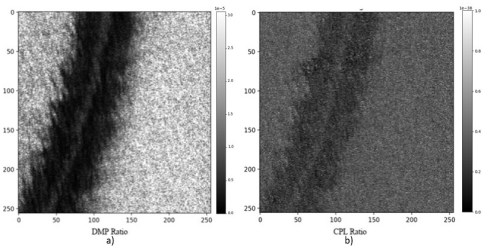


Fig. 11. Illustration of visual differences between VV damping ratio (DMP ratio, on the left-hand side) and co-polarized backscattering ratio (CPL ratio, on the right-hand side) for the mineral-oil-slicks class.

similar to the VV damping ratio, comparable in slick-sea contrast, shapes, and textural features, given the same dependence on large/small-scale roughness and acquisition geometry, and therefore, is not shown in consideration of space limitations. Fig. 10 shows the sensitivity to monomolecular biogenic streaklines in the WLD area, while Fig. 11 shows sensitivity to the thicker mineral oil films in the MC-20 area.

4) CNNs Opportunities in Oil Slicks Context: As discussed in Section III-B.3, radar backscattering is more sensitive to dielectric properties of thicker mineral oil films; hence, the co-polarized backscattering ratio products of thin films should be visually more similar to clean ocean surfaces, while thicker films should be visually more discernible from the clean ocean and comparable to the VV damping ratio and co-polarized difference counterparts. Ideally, when used jointly with both an ocean wave spectrum dependent (i.e., the VV damping ratio) and an ocean wave spectrum-independent feature with dielectric properties sensitivity (i.e., co-polarized backscattering ratio) as input, CNNs may learn features based on such differences that enable good separation between the three classes that are the object of this work (mineral oil slick, biogenic slicks, and clean seawater), e.g., at a high level of abstraction, CNNs might learn that thicker films appear to be similar in both products, while biogenic streaklines are less similar and clean-sea water surfaces relatively flat in both products. Furthermore, other information, such as slick shape and local statistics, may be learned in a hierarchical way, together with nonlinear features that can jointly help the classification. Additional considerations are discussed in the next sections, along with the presentation of experimental results and a demonstration of the opportunity to use CNNs in operational environments for slick classification.

IV. Results and Discussion

As outlined, an assessment of IRN and XCT CNNs architectures within a three-class (mineral oil slicks versus biogenic slicks versus clean-sea) classification problem has been performed in the present work. Single- and double-input evaluations have been considered, in order to investigate the best settings, including for possible operational scenarios. Based on theoretical considerations, single-input performances achieved using the VV damping ratio, and the co-polarized difference are expected to be much higher than the equivalent results obtained with the single-input co-polarized backscattering ratio since this latter should result in poorer classification potential due to limited sensitivity to small-scale roughness and to dielectric properties of monomolecular films, which mostly makes biogenic films appear comparable to clean-sea water and may cause biased learning effects. In contrast, damping effects observed with the VV channel have already been shown to be good indicators for detecting and characterizing slicks, and also for separating biogenic films from clean water and mineral oil slicks from clean water in [32]; furthermore, Espeseth *et al.* [32] and Angelliaume *et al.* [33] already proved that the PD is a robust feature for slick discrimination and for quantifying the oil-water mixing ratio. The possible use of statistical, textural, and shapes information on which CNNs are capable is expected to result in significant positive performance outcomes. Furthermore, the use of a double-input framework is expected to provide greater robustness since it takes more information into account from both input products by correlating recurrent patterns in the input data.

A. CNNs Settings and Environment

Tables V–XII show in detail the experimental outcomes achieved by testing CNNs in different situations; each network has been trained under equal conditions, with the same hyperparameters and without the use of data augmentation. In particular, all CNNs have been trained for 50 epochs, with batches size set to 32 and adaptive moment estimation algorithm (Adam) optimizer [89], which is based on the first- and second-moment estimations and has been demonstrated excellent for nonconvex optimization problems [89], such as training of deep networks potentially with multiple locally optimal points. All core operations, such as training, validation, and testing of the considered architectures, have been carried out by leveraging the TensorFlow open-source machine learning platform [90] with Keras interface (released with Apache and Massachusetts Institute of Technology (MIT) licenses, respectively), training CNNs from scratch by taking advantage of a highly parallelized GPU. The learning rate has been set to 10^{-3}, with default values for other parameters, such as ϵ, β_1, and β_2 [89].

B. Considered Metrics and Outcomes

Tables V–X indicate testing performances of XCT and IRN architectures with single-input configurations and both patches side length values (128 and 256) under study. Tables XI and XII show analogous testing performances of the two CNN architectures with double-input configurations.

Classification effectiveness is thoroughly examined by means of several metrics, namely, the conventional accuracy

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TABLE V

CNNs PERFORMANCES WITH SINGLE-INPUT VV DAMPING RATIO AND PATCHES-SIDE LENGTH 128

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.839	0.991	0.889	0.728	0.742	0.898	0.835
	30	0.838	0.925	0.994	0.708	0.883	0.789	0.821
	50	0.933	0.946	0.995	0.874	0.958	0.918	0.916
	70	0.964	0.937	0.984	0.975	0.962	0.946	0.974
IncResNet (IRN)								
	10	0.746	0.964	0.664	0.761	0.573	0.778	0.797
	30	0.788	0.964	0.981	0.626	0.857	0.725	0.759
	50	0.893	0.953	0.949	0.803	0.937	0.854	0.875
	70	0.864	0.938	0.951	0.758	0.894	0.839	0.839

TABLE VI

CNNs PERFORMANCES WITH SINGLE-INPUT COPOLARIZED BACKSCATTERING RATIO AND PATCHES-SIDE LENGTH 128

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.335	0.0	0.333	0.059	0.0	0.494	0.011
	30	0.333	0.0	0.0	0.333	0.0	0.0	0.495
	50	0.335	0.0	0.332	0.122	0.0	0.494	0.019
	70	0.334	0.0	0.0	0.334	0.0	0.0	0.496
IncResNet (IRN)								
	10	0.332	0.332	0.0	0.0	0.494	0.0	0.0
	30	0.333	0.0	0.333	0.0	0.0	0.493	0.0
	50	0.334	0.334	0.0	0.0	0.495	0.0	0.0
	70	0.334	0.334	0.0	0.0	0.497	0.0	0.0

TABLE VII

CNNs PERFORMANCES WITH SINGLE-INPUT VV DAMPING RATIO AND PATCHES-SIDE LENGTH 256

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.898	0.994	0.990	0.771	0.932	0.895	0.863
	30	0.982	0.982	0.990	0.976	0.987	0.981	0.975
	50	0.988	0.993	0.995	0.978	0.993	0.989	0.982
	70	0.897	0.980	0.967	0.787	0.909	0.908	0.868
IncResNet (IRN)								
	10	0.843	0.990	0.983	0.690	0.948	0.749	0.808
	30	0.911	0.992	0.976	0.810	0.930	0.912	0.887
	50	0.898	0.975	0.951	0.808	0.920	0.881	0.885
	70	0.953	0.954	0.993	0.922	0.938	0.971	0.945

TABLE VIII

CNNs PERFORMANCES WITH SINGLE-INPUT CO-POLARIZED BACKSCATTERING RATIO AND PATCHES-SIDE LENGTH 256

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.337	0.335	0.0	0.114	0.495	0.0	0.021
	30	0.338	0.335	0.0	0.138	0.496	0.0	0.025
	50	0.338	0.335	0.0	0.163	0.495	0.0	0.029
	70	0.334	0.334	0.0	0.0	0.494	0.0	0.0
IncResNet (IRN)								
	10	0.333	0.0	0.0	0.333	0.0	0.0	0.494
	30	0.332	0.0	0.0	0.332	0.0	0.0	0.493
	50	0.331	0.331	0.0	0.0	0.491	0.0	0.0
	70	0.331	0.331	0.0	0.0	0.492	0.0	0.0

TABLE IX

CNNs PERFORMANCES WITH SINGLE-INPUT CO-POLARIZED DIFFERENCE AND PATCHES-SIDE LENGTH 128

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.982	0.981	0.994	0.972	0.983	0.988	0.972
	30	0.982	0.978	0.986	0.982	0.981	0.990	0.972
	50	0.992	0.992	0.997	0.989	0.993	0.995	0.989
	70	0.982	0.976	0.982	0.980	0.981	0.986	0.970
IncResNet (IRN)								
	10	0.979	0.980	0.990	0.966	0.980	0.986	0.966
	30	0.986	0.985	0.990	0.977	0.985	0.991	0.975
	50	0.987	0.994	0.996	0.971	0.986	0.994	0.979
	70	0.979	0.962	0.983	0.992	0.973	0.988	0.975

TABLE X

CNNs PERFORMANCES WITH SINGLE-INPUT CO-POLARIZED DIFFERENCE AND PATCHES-SIDE LENGTH 256

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.991	0.987	0.998	0.989	0.989	0.996	0.987
	30	0.987	0.984	0.995	0.982	0.990	0.990	0.977
	50	0.979	0.981	0.989	0.963	0.979	0.981	0.964
	70	0.989	0.986	0.988	0.991	0.987	0.991	0.984
IncResNet (IRN)								
	10	0.986	0.983	0.988	0.985	0.986	0.989	0.977
	30	0.987	0.987	0.990	0.976	0.987	0.991	0.973
	50	0.981	0.984	0.990	0.974	0.987	0.983	0.972
	70	0.921	0.943	0.956	0.835	0.877	0.968	0.869

TABLE XI

CNNs PERFORMANCES WITH DOUBLE-INPUT AND PATCHES-SIDE LENGTH 128

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.911	0.982	0.974	0.807	0.886	0.953	0.880
	30	0.920	0.841	0.990	0.942	0.907	0.971	0.867
	50	0.976	0.978	0.979	0.968	0.982	0.969	0.973
	70	0.934	0.955	0.981	0.881	0.975	0.899	0.922
IncResNet (IRN)								
	10	0.722	0.715	0.712	0.771	0.787	0.796	0.468
	30	0.825	0.962	0.944	0.683	0.775	0.890	0.794
	50	0.889	0.925	0.996	0.797	0.945	0.823	0.877
	70	0.869	0.949	0.898	0.783	0.937	0.785	0.857

TABLE XII

CNNs PERFORMANCES WITH DOUBLE-INPUT AND PATCHES-SIDE LENGTH 256

Architecture Xception (XCT)	Oil (%)	Acc	Prc WX	Prc TL	Prc OC	F1 WX	F1 TL	F1 OC
	10	0.942	0.980	0.984	0.873	0.969	0.932	0.916
	30	0.919	0.954	0.995	0.830	0.951	0.899	0.891
	50	0.918	0.930	0.995	0.865	0.957	0.876	0.910
	70	0.961	0.963	0.975	0.949	0.959	0.974	0.949
IncResNet (IRN)								
	10	0.887	0.983	0.887	0.819	0.904	0.883	0.861
	30	0.977	0.979	0.988	0.963	0.982	0.979	0.967
	50	0.894	0.968	0.992	0.776	0.932	0.869	0.866
	70	0.867	0.850	0.969	0.815	0.897	0.797	0.867

metric, precision, recall, and F-1 score; since the latter three have mutual dependence, only the precision and F-1 score are shown for summary purposes. All of these different metrics are defined in terms of true/false positives (TP/FP) and true/false negatives (TN/FN) [88]. Accuracy is defined in terms of the ratio between correctly classified samples to the total number of samples

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{11}$$

Recall is defined in terms of the number of correctly classified positive samples to the total number of positive samples

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{12}$$

Precision is defined in terms of the positive correctly classified samples to the total number of positive predicted samples

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{13}$$

F-β score is defined as a weighted harmonic mean of precision and recall; accordingly, the F-1 score represents the harmonic mean, and higher values suggest higher classification performances

$$F_1\text{Score} = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}} \tag{14}$$

Results presented in Tables V–XII are visually summarized in Fig. 12(a)–(d), from which general conclusions can be derived.

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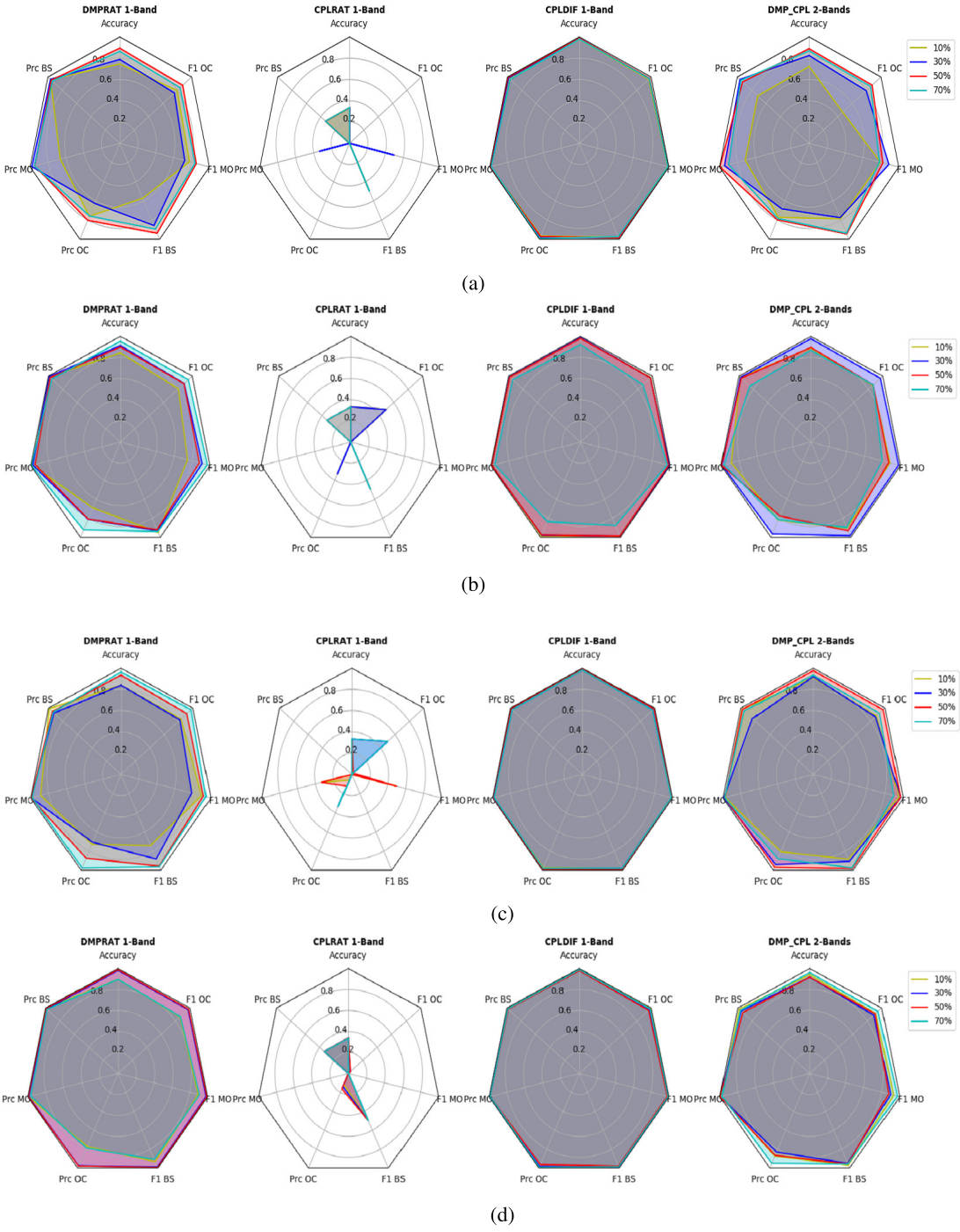


Fig. 12. Graphical presentation of experimental results, in terms of the metrics described by (11)–(14). Prc and F1 are abbreviations for precision and F1-score metrics, while DMPRAT and CPLRAT, respectively, refer to VV damping ratio and co-polarized backscattering ratio. Different oil-occurrence percentages correspond to separate colored lines, while BS, OC, and MO are acronyms for the biogenic-slick, clean-ocean, and mineral-oil classification categories, respectively. From left to right, single-input performances are shown first, while the last plot shows double-input performances. (a) IRN performances, with patches-side size 128. (b) IRN performances, with patches-side size 256. (c) XCT performances, with patches-side size 128. (d) XCT performances, with patches-side size 256.

An interesting result involves the double-input application, which is clearly shown as generally improving upon the VV damping ratio single-input performances. As predicted by theory, it is confirmed that the exclusive use of the co-polarized backscattering ratio leads to poor classification capability, while the exclusive use of VV damping ratio and polarized difference products proves them to be reliable parameters for characterizing oil slicks, as shown in [32] and [33], and both may constitute a reliable classification tool when equipped with CNNs capabilities of dealing with shapes, textural and statistical patterns, even when only single-polarization sensors are available in the case of VV damping ratio. From an operational perspective, the XCT architecture with double-input configuration and patches size 128 appears

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to be fairly robust in any oil-occurrence case and may represent a suitable configuration in operating mode. Highest scores, in absolute terms, are obtained from the XCT architecture with single-input (VV damping ratio) configuration and patches size 256, and single-input (polarized difference) configuration and patches size 128, both when trained/tested over the 50% oil-occurrence data set [see Tables VII and IX and Fig. 12(c) and (d)]. In comparison, the same configurations over the 30% oil-occurrence data set show slightly different outcomes, although this latter can be considered even more valuable and robust given that 30% oil-occurrence data set approximately contains double the number of samples of the 50% oil-occurrence data set.

C. Best Case Scenario in Operational Environments

The most probable condition in real-world scenarios is likely represented by the 10% oil-occurrence situation since deployed sensors are typically designed with wide to very-wide swaths, and general acquisitions are not always well focused/centered on slicks areas. Consequently, XCT architecture scores with double-input and single-polarized difference input configurations, and both patches side size 256 and 128, represent the most important result, suggesting ideal configurations for future studies and operational applications [Tables IX–XII, Fig. 12(c) and (d), right panels]. This architecture also indicates an accurate method of separating the three classes of mineral oil slicks, biogenic slicks, and clean water.

V. Conclusion

An assessment of deep CNNs' capabilities in oil slick classification, discriminating between mineral oil slicks, biogenic slicks, and clean seawater, with different configurations and data set parameters, has been presented. In particular, a three-class classification problem is set, with high classification performances achieved by the two investigated CNNs architectures, namely, XCT and IRN. A few authors have claimed oil slick discrimination may be possible based on scattering mechanisms analysis since non-Bragg behaviors can be observed in the presence of mineral oil slicks, while Bragg scattering applies in other conditions; however, convincing arguments against the occurrence of non-Bragg scattering mechanisms and evidence of compliance with tilted Bragg scattering model have been published. This suggests that non-Bragg results at low-to-moderate wind speeds are correlated with noise-contaminated data; thus, products with the low-noise floor are recommended for oil discrimination studies. In this regard, the study's use of data from UAVSAR, a sensor with an extremely low-noise floor, enables an uncontaminated analysis better, and therefore, it has been selected as the reference instrument in this work.

A. Relevance of Inputs Choice and Final Remarks

The use of different types of data inputs, namely, VV damping ratio, PD, and co-polarized backscattering ratio, was selected based on achieved significant positive results, even for relatively low slick coverage within a patch, which is likely to be observed in operational oil spill monitoring with spaceborne SAR. Even though various features seem to be not suitable for proper discrimination, such as the considered use of Bragg/non-Bragg scattering mechanisms, the combined use of a powerful tool for data statistics evaluation, including spatial patterns (CNNs) and data products that are sensitive to small-scale roughness, dielectric, and damping properties of slicks, is proven to have highly accurate results. Multiple configurations have been discussed and analyzed in this work, including the use of specific data sets with various patch size and oil-occurrence percentages. From the obtained results, it is evident that the use of XCT architecture, with double-input configuration (i.e., the use of both the damping ratio and the co-polarization ratio) and single-polarized difference input configurations, and both patches side length 256 and 128, represents an optimal setting. This is particularly true for discrimination purposes in a possible operational situation, having shown very high outcomes, specifically accuracy values of 0.91, 0.94, 0.98, and 0.99 for the 10% oil-occurrence case, which represents the most probable condition in real-world spill scenarios. Although the results obtained already represent a significant milestone, also given that the large data set employed in our experiment makes the obtained results very robust and reliable, the present work also serves as a baseline to demonstrate the high oil slicks classification potential of CNNs with SAR data and to foster further and different assessments (e.g., of different CNNs architectures, with the use of data-augmentation, with even larger data sets) that may lead to even better findings.

Acknowledgment

UAVSAR data courtesy NASA/JPL-Caltech. This research was carried out in part at the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration. UAVSAR data can be downloaded from uavsar.jpl.nasa.gov or from the Alaska Satellite Facility (http://www.asf.alaska.edu).

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